

Asset Share

- Widely used for the management of with profit business but can also be determined for NP.
- Asset share is the retrospective calculation, i.e., its value is totally determined by looking at the historical cashflows that have already occurred. .
- In case, the PH decides to surrender the policy, then company calculates their asset share and may not pay them exactly 100% of it. This include addition

If there are n identical policies over the next year, the aggregate Asset share at time t+1:

$$n(AS_t + P + E_t)(1 + i_t) - Snq_{x+t}^n$$

Asset share for individual policy =

$$AS_{t+1} = \frac{n(AS_t + P + E_t)(1 + i_t) - Snq_{x+t}^n}{n(1 - q_{x+t}^n)} = \frac{(AS_t + P + E_t)(1 + i_t) - Sq_{x+t}^n}{p_{x+t}^n}$$

Where P – annual premium

E_t = per-policy expense

I_t – investment return in year t to t+1

S- death benefit

q_(x+t)^" – actual mortality rate in the year t to t+1

Where does the money comes from for NB Strain?

- Surplus/profit generated by older policyholders from previous generations that was held back (undistributed) specifically to serve as working capital.
- Shareholders free assets, i.e., Capital invested by the owners of the company.

Asset Share calculations

1. WP regular premium endowment assurance on 1st Jan 2015. Premium=1000 annually and SA= 15000

Year	2015	2016	2017
IR	12.6%	8.5%	7.9%
Mortality	0.0014	0.0015	0.0017
Exp	1,320	84	89

Other profit – reduction in the yield of 0.3pa

Asset share at end of 2017

= We know, $AS_{t+1} = \frac{(AS_t + P + E_t)(1 + i_t) - Sq_{x+t}}{(1 - q_{x+t})}$, then

$$AS_1 = \frac{(AS_0 + P + E_0)(1 + i_1) - Sq_1}{(1 - q_1)} = \frac{(0 + 1000 - 1320)(1 + 0.123) - 15000(0.0014)}{(1 - 0.0014)} = -380.8932$$

$$\text{Similarly } AS_2 = \frac{(-380.8932 + 1000 - 84)(1 + 0.082) - 15000(0.0015)}{(1 - 0.0015)} = 557.325$$

$$AS_3 = \frac{(557.325 + 1000 - 89)(1 + 0.076) - 15000(0.0017)}{(1 - 0.0017)} =$$

Asset share development

Expense – heavy initial expense and then much lighter subsequent to that

Cost of life cover – increase year by year

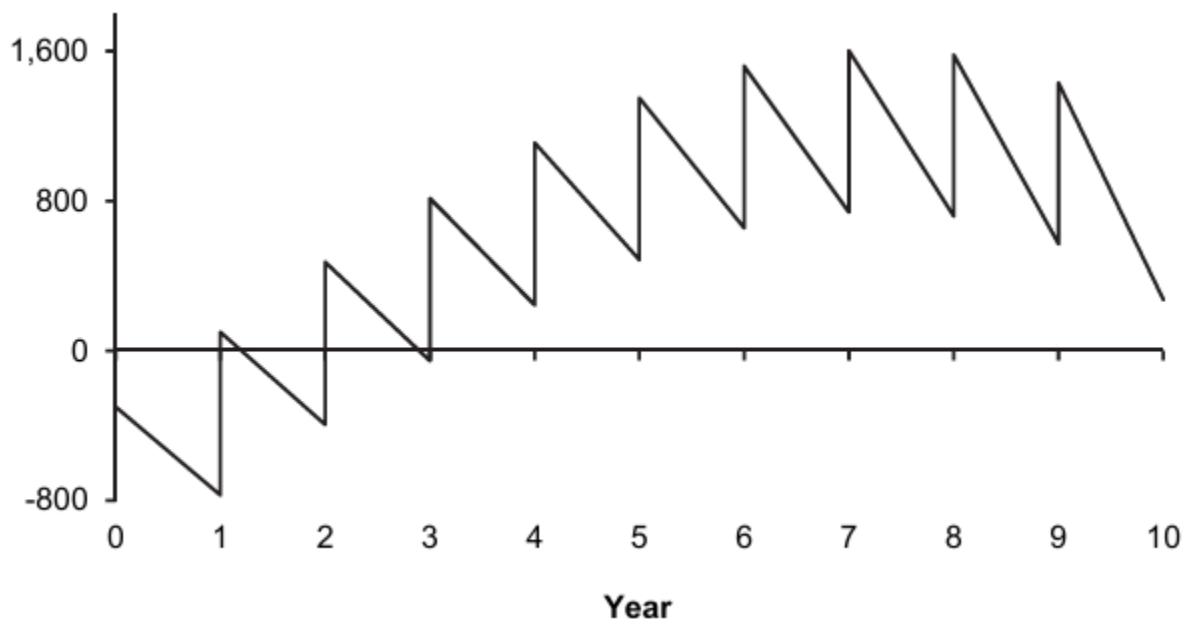
Investment income – compounding effect unless deduction due to SH transfer are particularly big

At maturity $AS > \text{Guaranteed SA}$. Difference is the profit and is distributed to SH or PH-

How the AS moves after inception will primarily depend on –

- Component of renewal expense
- Investment income
- Life cover all balance out

For regular premium term assurance – Asset share graph looks like below



- The steep falls represent the cost of death claims, rise at each policy anniversary
- AS is initially -ve because each year's premium > cost of life cover in the early years.
- Then, as the PH ages Cost of cover > level of premiums and the AS decreases

- At the end of the policy's term the asset share is small and positive. There is no maturity benefit to provide, but the company should be making some profit on the contract.

Reasons for asset share to reduce

- Asset share is the accumulation of premium less deduction accumulated at the actual IR.
- So reduction indicates that deduction are higher than premiums or an IR have been negative.
- Equities and/or property values could have fallen, or interest rates could have increased reducing bond values
- Premiums could also have reduced due to policies being made paid up.
- Expenses and Mortality could have been high especially at early duration
- Tax could have been high
- $SV > \text{asset share}$
- If aggregate asset have been reduced, then number of policies may have reduced due to: business being run off; a higher number of maturities, surrender and death; a reduction in new business
- There may have been an error in the calculation
- The asset share methodology may have changed, eg using actual instead of smoothed investment return.

Actions to take following asset share reduction

- Check the calculation for errors.
- If individual asset shares have reduced, company should determine the cause
- If expenses have been high, investigate ways to reduce.
- If surrender payout is high then try to change that
- If there are losses on other business then investigate the cause and take proper action
- If IR is negative then this should be compared with overall market return.
- If asset shares have fallen to a point where guarantees could bite, then guarantees need to be closely matched.
- Depending on the driver for reduction, consider reducing reversionary bonus or more likely terminal bonus.
- Consider adjusting the smoothing approach.
- Actions will depend on the - PH expectations, the need to treat all the PH fairly, the need to be inline with the regulation, competitor practice if the cause is market driven.

Impact of following on WP whole life regular premium policy

- Surrender of PH – Depends on how surrendering PH were treated – If $SV = AS - \text{Cost}$ then the AS of remaining PH will be unaffected. If $SV < \text{Cost}$ then higher AS. SV should not be bigger than AS because it will cause loss equal to the amount of excess to the company
- MV of asset down by 10% - down by 10%
- SH rate of return Increases – increase cost of the SH, share of distributed profit increase. Increasing the amount of SH transfer $\Rightarrow$ dec in AS
- Reserves increased – no immediate effect. But other effects could be –
Strengthening the reserving basis – policies will become more capital required to sell a policy will not be higher. $\Rightarrow$ deduction from AS at early durations $\Rightarrow$ reduce future AD

- Reduce the investment freedom – reduce the investment return -> lower AS
- v) Policyholder dies – goes down. However one death will have no influence.

Impact of SV > asset share

- Company would make loss and this loss would be passed onto other PH asset share as a deduction.
- Bonus distribution should be equitable between different categories and generation of PH and paying out more to surrendering PH would not be equitable to those remaining.
- If PH were made aware of the lack of equity between surrendering and continuing PH, then more might surrender meaning even more deduction to those remaining.
- This would not be sustainable, and so either SV will reduce or SH would make a loss.
- Bonus distribution should be consistent with PHs reasonable expectations, eg SV should be inline with the AS.
- In the early years of the policy the asset share could be negative due to high initial expenses. As it is not possible to have a negative SV, any SV would then be technically exceed AS

With Profit surplus distribution (1)

Profit is referred as 'surplus or surplus arising'

Possible ways of distributing profit to WP PH –

Main possibilities are – cash bonus, premium reduction, benefit increase

1. Life insurance company gives bonus as cash now want to change to benefit increase. Possible constraints

⇒ -Mentioned below

- Legislation – legal requirement prevents
- Policyholder expectation – will not be accepting this approach. May not appeal to the company's normal clientele and inability to win them over may cause marketability risk
- Competition
- System limitation – may not be able to code with different data required for the new sys
- Management limitation – training and expertise

Deffering the distribution of profits –

- Probability of remaining solvent increase by reducing and delaying the distribution of profits
- Investment freedom

2. How delaying reduce probability of insolvency

⇒ Superficial level – may seem obvious that if it retains profit then current level of free assets increases.

However on deeper level –

- If delaying means that the amount of profit needs to be paid out at some future date, then it becomes a fixed liability for the company and is then unlikely to improve the company's solvency position.

- If delaying means deferring the decision on how much it is going to pay out, then the company transforms a rigid liability into a flexible one, i.e., company should eventually payout something in respect of those retained profits but the monetary amount can be reduced if the profits have fallen in value by the time they have to paid
3. 'Surely PH will receive the bonus later, so of course the distribution is delayed'.
- The delay here is in the payment of the profit amount. But profit is distributed the moment a company announces the amount of profit to be paid. Because then the company has made a promise to pay the amount on the policy claim date. So no matter what happens to the asset share between the bonus announcement date and the date of claim payment, the insurer is obliged to pay it.

So, how can underpaying people today lead to giving them more money overall in the long run?

The answer mainly lies in terminal profit distribution, all the profits we retain from underpaying bonuses throughout the policy term, we pay back in a big lump (hopefully) as an extra distribution on the day the policy terminates. The higher returns that we hope to have made in the meantime should result in higher terminal distributions being paid, at least in the long term

However, this makes marketing difficult.

Meeting policyholder's needs

Potential Reduced benefit of policy due to cash bonus or premium reduction instead of terminal bonus can be unfortunate for the PH, if the maturity amount is not able to meet the eventual need of the PH.

Situations where paying cash dividends as bonus is suitable –

- Has a large amount of assets to invest now but has little future income - single premium WP. Cash bonus would meet the need of income and survival or death benefit would protect the capital
- person with fixed monetary liability at future date – regular premium WP endowment with the sum assured = loan amount

Distribution of profit

Here Profit is

- Excess of investment return on premium paid, over and above the guaranteed return.
- Better mortality or expense experience
- From other products. Whether or not WP PH are entitled to profits made on this business depends on each company's situation.

Addition to benefit approach

- Profits are distributed in direct relation to the current benefit under each contract.
- Conventional additions to benefit approach are the system of distributing profits to PH, using reversionary and terminal bonuses,
- In this the bonus added to each policy is defined as some proportion of the benefits currently payable on a claim under that policy.

- The bonuses are then added to the contractual benefits payable under each contract on maturity or death
- Under this method bonuses are added using three kinds –
 - o Regular revisionary bonuses, added throughout the contract term
 - o A special revisionary bonuses, added as a 'one-off' from time to time
 - o Terminal bonus, when contract ends due to death, maturity or surrender.

Determining regular reversionary bonus rates

- Declared regularly throughout the term of contract.
- Once declared it is attached to benefit.
- Do not vary greatly from one year to another.
- They change gradually over the period but not suddenly.
- Since market conditions vary, the company estimates the level of regular bonus it could afford to continue paying into the future according to the assessment of future experience.
- Therefore, bonus earning capacity of a block of contract is the rate or rates of bonuses that that contract can sustain over their future lifetime, based on a set of assumptions with regard to future experience.
- Bonuses declared in a year should also be affordable in the short term.
- The company will look at its current bonus rates, check if this differs from the long-term sustainable rate, and may consider changing it to the nearer LT Rate.
- The progression of the company's regular bonus rate over the last few years would also be considered when considering the rate to declare for the current year.
- The amount of bonus can be calculated in one of the three ways:
 - simple - bonus is expressed as a percentage of the basic benefit under the contract
 - Compound - the bonus is expressed as a percentage of the basic benefit plus any already attached bonuses
 - Super-compound - the bonus is expressed in terms of two percentage: one applied to the basic benefit and a second applied to any already attached bonus. The second percentage is typically higher than the first. This approach means, later distribution of bonus=> lower build up of guarantees=> investment freedom=> reduces the risk of insolvency=>more marketable=>could attract different PH then existing=> increase in market share. Dis - regular bonus may look small compared to simple, may look less attractive=>complex=> additional explanation to PH => needs expertise other wise risk in incorrect bonus distribution => may not seen as equitable =>may require regulatory approval

Special reversionary Bonuses

- This could happen as a result of restructuring a with-profit fund.
- This is done to grant a one-off increase without creating expectation that a similar increase will be granted in the future.

- This is given for any special circumstances that are unlikely to be repeated may arise from demutualization (member owned mutual organization converts into publicly traded stock company) or takeover

Terminal Bonuses

- Defers the distribution of surplus until the end of the contract.
- Slows down the build up of guarantees.
- Useful for distribution of profit that came from volatile sources
- Determined when the insured event happens.
- Payout is generally equivalent to the asset share of the policy.
- Terminal bonus=AS-Guaranteed benefit till date (SA+Reversionary bonus+Special reversionary bonus(if applicable))
- Companies will then 'smooth' the payout to reduce the impact of fluctuations.
- Implying that terminal bonus can be more or less than that needed to bring the payout exactly up to the asset share.
- Ways of terminal bonus
 - o % to reversionary and special reversionary bonuses. %age may vary depending on the policy duration in force
 - o % of total claim amount before terminal bonus. %age will vary depending on the policy duration in force

4. Why for first it would possibly vary and for second it would definitely vary?

- ⇒ Because given a reasonable smooth, non-volatile investment return, the undistributed surplus should be approximately the same proportion of the distributed surplus of all duration.
- ⇒ Therefore terminal bonus as a fixed %age of revisionary bonus may bring the payout reasonably close to the asset share of the policies of varying terms.

5. Why do we need reversionary bonus and terminal bonus both?

- ⇒ PH likes reversionary bonus. Without that they may not feel that they are accruing any profit and their financial planning will be difficult.

And terminal bonus increases company's investment freedom allowing investment in high risk/high return assets. So the eventual payout should be greater.

Terminal bonus also allows smoothing which would be impossible to do at a significant level with only reversionary bonuses.

Accumulating with Profits

- This is more transparent in nature
- Individual Policyholder's account is maintained by the insurer which is also visible to the PH.
- Bonuses are added annually in relation to the premiums payable to date plus previously declared bonuses.
- When the insured event happens, company may add a terminal bonus to the bid price of the units and may also add special bonuses in duration inforce.

Difference between accumulating WP and Conventional WP

- In Conventional WP No immediate obvious relationship between the stated/ benefit and any present value of the policy
- AWP is more like a bank savings account
- AWP – Account starts from 0 and is increased by the amount of premium paid and by bonuses.
- Payout in CWP – Sum assured + Reversionary bonus + Terminate Bonus
- Payout in AWP – (Premium + declared bonus)
- CWP minimum sum assured is always available. However, in AWP, a minimum sum assured might be offered but its rare.
- Guarantees in CWP are likely to be way greater than those provided by AWP because in AWP, guarantees are declared on Current account balance rather than SA.

There are 2 ways of distributing the surplus in this type of policy

- Unitized WP and
- **Non-Unitized WP** – Very much like conventional WP, difference is explicit relationship between each single premium paid and the additional benefit

	CWP	Non-Unitised WP
Mechanism	Policy starts with Guaranteed SA. Bonuses are declared annually as a percentage and are given by three means, i.e., cash bonus, premium reduction, addition to benefit	Maintain transparent nominal cash account balance, i.e., visible to policyholders. Every year bonuses are declared as a monetary interest rate which is credited directly to account balance, causing it grow
Transparency	Policyholder only knows how their total guaranteed SA payable on death or maturity	An online account is maintained where they can see how much cash is sitting in their account at any given moment just like a bank account.
Value	SA + accumulated RB + Terminal bonus	Cash account balance. (Premium - expense) (1 + surplus rate)
Regular Bonus impact	Increases eventual payout at death or maturity	Directly increase monetary value of the cash account via an interest rate credit
Surrender Value Calculation	% of asset share or formulaic perspective	Cash account balance – surrender penalty

$$F_0 = 0$$

$$F_1 = (F_0 + P_0 - c_0)(1 + b_0) = (P_0 - c_0)(1 + b_0)$$

$$F_2 = (F_1 + P_1 - c_1)(1 + b_1)$$

$$F_{t+1} = (F_t + P_t - c_t)(1 + b_t)$$

Where:

F_t = policy benefit (excluding terminal bonus) at policy time t ; $F_0 = 0$

P_t = premium paid at policy time t

c_t = charge deducted at time t

b_t = regular bonus added between times t and $t + 1$

Unitized WP –

premium is converted in terms of units. Benefit amount is = value of units

The two basic ways in which the unit part of the contract could operate –

- The price of a unit remains constant and additional units are allocated as bonuses. These number of additional units are determined at the discretion of the company.
- The price of a unit changes and the number of units increases. Increase is made up of guaranteed part and a bonus part

Balancing mechanism (MVR and terminal bonus)- because UP is smoothed, it drifts away from the actual value of the underlying assets.

- If policyholder exists when market is booming – terminal bonus is given
- If they exit during a market crash, the insurer applies Market value reduction (MVR) to bring back the price within the asset share. Calculated at the discretion of the company. This is used because bonuses are smoothed and even after the surrender penalty deduction, unit value could be above the value of the underlying asset share. In such circumstances, the company will be vulnerable to surrenders.

This can also be applied to non-unitised WP.

Charges of Unitized WP are similar to unit linked –

- Policy charge or policy fee – can be taken from premium or the fund
- Percentage allocation during an initial period
- A different percentage allocation after the initial period
- Bid offer spread
- Charge of risk benefit
- Annual management charge

These charges could be taken through the bonus rate, with no explicit charging structure

	Unitized accumulation WP	Unit linked
Mechanism	Does not mirror daily market movements. It grows steady via smoothed bonus declaration. Once these bonuses are added they can't be taken away	Unit price are tied to underlying investment fund.
Smoothed return	Yes	no
Investment risk	Shared – guarantees minimum growth rate – taking on the downside risk of a market crash	100% borne by PH
Downside volatility	Low at the PH, but due to smoothing bonus, company have to bear	High at the PH
Addition or reduction in benefit	Terminal bonus and MVR applicable on surrender or discretionary exit	No such reduction or addition to the amount of unit linked

Guarantees	Low, the only guarantee is unit price will never decrease	High, in case minimum SA is guaranteed in case of maturity or death
Transparency	Unit price reflect smoothing choices and not raw asset value	Exact value of market
Capital requirement	High, to smooth the return whenever market performs badly.	Low, when policy has aged

6. UWP – annual premium – 8K. Guaranteed DB – 200000. 102% of the premium is allocated to the units. Charges – bid offer spread – 4% , policy fee – 60, management charges – 0.5%, charge for risk benefit – sum at risk at the start of year*0.004

Value at start of the year – 34500 before the payment of premium.

Bonus = 5% calculate the price

I) Calculate the value at the end

$$\Rightarrow 34500 + (8000 * 0.96 * 1.02) - 60 = 42273.6$$

$$\text{Value at risk} = 200000 - 42273.6 = 157726.4$$

$$\text{Risk charge} = 157726.4 * 0.004 = 630.9056$$

$$F = (42273.6 - 630.9056) * 1.05 * 0.995 = 43506.205$$

II) Terminal bonus – 15%. What amount would be paid to PH –

$$= 200000$$

III) Surrender benefit > MVR = 3% and surrender penalty = 750

$$SV = 43506.205 * 0.97 - 750 = 41451.019$$

Considerations while bonus distributions

- Consistent with PH expectations
- Be equitable between different categories (policy type, term, size, sex, territory) and generation of PH
- Inline with future business plans, investment strategies or solvency of the company – when we declare bonus, we increase the guarantees under all the policies and insolvency risk increases. Therefore writing large amount of bonus is risk. Declaring large bonuses could also restrict and reduce the available capital for other ventures.

Asset share and bonus distribution

- Asset share – amount insurance company has for WP policies
- Thus can be used as a tool to determine reasonable and equitable payout to different generations and classes of WP PH.

7. Why asset shares are used for determining payouts for WP

$\Rightarrow$ Could calculate AS for UL and conventional NP contracts. However, benefit of UL is calculated by the size of the unit account in conjunction with any guarantees and for NP – guaranteed SA. Thus no need to calculate AS. The only use of AS would be to monitor profitability once the contract has reached maturity by comparing asset vs benefit paid. It can also be used to limit the calculation of SV.

Reasons why AS and payout from WP policy could be different.

- AS could be less than guaranteed amount.
- Benefits may have been smoothed, so it could be higher or lower than the unsmoothed AS.

- The benefits may have been set by reference to an average or sample policy that differs from the individual policy in question.
- Payouts may be net of any PH tax due.
- The payout of surrender may be different to AS because: a simpler approach is used, a prospective valuation method is used, there is a surrender penalty, the policy is surrendered early in term when the AS is negative, an alternative approach is used when AS is low.
- Company may distribute its provide by method other than asset share eg - revalorisation method or demutualisation bonus.
- Actual investment return on asset share is different than the assumed during the bonus rate calculations.
- There may be lack of data which leads to a need to approximate the AS.
- There may be error in the payout calculation that is causing the difference.

With Profit surplus distribution (2)

Revalorisation Method

- Used in parts of Europe, eg, Italy, France
- Profit to be given to a particular contract is expressed as $r\%$ of the contracts supervisory reserve
- The benefit and premium payable by PH are then increased by same amount.
- Premiums are increased by the same proportion because otherwise company won't be able to afford increase to the sum assured

$$\text{Reserve } V = S\bar{A} - P\ddot{a}, \text{ hence } (1+r)S\bar{A} - (1+r)P\ddot{a} \quad A\bar{\mid} \text{ and } a\ddot{\mid}$$
- Then if premium has also gone up, how is this bonus? Because this increase in SA is based on past premiums paid and future premiums will be increased.
- Most countries do offer a constant premium policy, however benefit will be less for those.
- Companies split their profits into 'savings' profit (generated purely from the assets) and 'insurance profit' (better mortality or other assumption experiences)
- Savings profit can be distributed in whole or in part at the end of policy year. If a policy surrenders or claims before then, some pro rata adjustment is made.
- r represents the excess of actual investment earned, $r=i'-i$ (i' actual IR and i – expected)
- So companies usually use the formula, $r=k(i'-i)$ or $r=(k.i'-i)$ where $0.75 \leq k \leq 0.95$ and $r \geq 0$
- Calculation of actual returns will vary from country to country – derived from investment income, realized GnL, and unrealized GnL
- Insurance profit is generally retained by the company for distribution to SH, as reward for the pure insurance risks that they have borne. Sometimes can be divided in PH and SH.

Advantages

- Simple to apply
- Confides how company should declare part of its profit as bonus to WP PH. Therefore cheap to administer.
- Protects PH from ungenerous LI companies
- By taking assets at Book value (price insurance company paid to buy assets), including appropriately smoothed writing-up adjustments (smooth increases in the value of asset yearly so

that a sudden surplus doesn't arise only when asset is sold), a smooth emergence of investment profit is usually achieved.

Disadvantages

- No discretion in profit distribution
 - Discourage equity investment because of no deferral of profit distribution.
 - Implying all investment losses will be borne by company and would constitute an acceptable insolvency risk.
 - Unrealized gains are not easy to distribute directly under the current revalorization system.
 - Versions that do not share profit with PH go against principle of mutuality (principle of mutuality dictates that – PH join together to pool their risk – contribute premium to a common fund – because they own the risk collectively therefore any surplus or profit generated by pool belongs entirely to the pool participants (the PH), not to external corporate SH.)
 - Not easy to explain with 'constant premium' policies who see very small addition to their guarantees benefit early in the policy term.
1. Difference in the way the revalorization method works for the mutual life insurance company (insurance provider that is completely owned by its PH).
- ⇒ Insurance profit would need to be added to the saving profit
 This will be expressed as a percentage of reserves and then passed on to the PH by increasing premiums and benefit in the same proportion, as usual.
 The increased proportion will be much closer to 100%, because there would be no SH with whom to share profit

Contribution method (US)

'the contribution principle' – surplus should be distributed in the same proportion as those judged to have contributed to surplus which makes it an extremely equitable profit distribution.

It can be calculated as

$$Dividend = (V_0 + P)(i'' - i) + (q - q'')(S - V_1) + E(1 + i) - E''(1 + i'')$$

Where

Vo	Value of contract at the beginning of year on valuation basis
V_1	Value of contract at the end of year on valuation basis
P	Premium received
i''	Actual rate of return earned
I	Valuation basis interest
I''-i	Excess of return
(q-q'')	Difference between the actual and expected mortality experience in the valuation basis
E''	Actual expense incurred
E	Expected expense in the valuation basis

The purpose of the formula is to divide the total amount of distributable surplus between the PH as *fair* as possible, in accordance to policy's contribution to the company's investment, mortality and loading surplus.

- The proportion of surplus will be smoothed to obtain more stable progression of dividends over time.
- Some profit will be retained for SH, held against future adversaries, or for terminal dividends.
- Unlike UK, it is quite common for dividends to be paid out as cash sum, or to reduce future premiums. However, paid-up addition to the benefit is also quite common.
- Which makes it like UK reversionary bonus system.
- The group of policyholders is generally more homogeneous in nature.
- In US the approach leads to much more of an attempt to give surplus back to specific group of PH than in the case of UK.
- For example, US and UK both allow different mortality at different ages. In the UK, that is pretty much the end of the story, i.e., it is highly unlikely that different bonus is paid to different age group no matter what the mortality experience of it. However, in US, surplus +ve mortality contribution will be given to age group which had better mortality experience.
- A terminal dividend may be given which could be substantial.
- Because when the contract goes off the book, the contingency provisions are no longer required.
- The company may retain a part, on the grounds that the PH has benefited from the protection and support of capital built up from earlier generations of PH. Therefore he/she should also contribute for the protection of future generations.

2. Suggest one theoretical and one practical reason why these policies will be grouped?

⇒ If we treat every policy individually we will lose the concept of pooling of risk. For example, PH who survived the year will show mortality profit and who didn't will show huge loss, and thus we use the idea of actual experience being the average experience of all policies in a homogeneous group.

Practical reason – to distribute profits perfectly and fairly, we would want to ground PH into highly specific tiny categories based group but then we will end up with thousands of different cohorts. We also need to ensure that there is sufficient data in each group to make the experience credible.

	PH	Insurer
Adv	bonus is received straight away instead of when the insured event arises, which may not be inline with the needs of PH. The bonus may appear fair to PH	if attractive to PH, will boost sale – marketed as new innovation
Dis	– reduced return due to limited deferral of distribution of surplus – may want reduction in premium to save fixed amount each month – final benefit is lower in real terms	reduce the level of funds under management and so may place constraints on investment policy may seem low in relation to reversionary it reduces company's investment freedom affecting overall returns Administration of cash bonuses would be more difficult and more expensive than an addition to benefits

Choice of with-profit bonus methods

3. Compare all three systems

	Additions to benefits	Revalorization	Contribution
Equitable and smoothing	Deliberate smoothing of asset share volatility over time. Sharing of mortality and expense surplus is only very broadly equitable	Smoothing over time is achieved by valuation method Equitable distribution of investment profit. Mortality and expense surplus is not included	Some smoothing Very equitable regards to insurance and saving profit.
Flexibility (discretion)	Max to insurer on how much bonuses to give and when	No discretion. Apply the set formula	Considerable flexibility. Total is split fairly by fixed rules, although some judgement as to groupings for application of the formula
Simplicity	Simple to tell PH on what company is doing but difficult to explain. They need to trust the industry	Simple	Complex to apply and explain. But easier than UK
Investment freedom	Lots, especially if conventional, use high TB and super compound RB	Very little, surplus is distributed as it arises	Immediate but some surplus is held back for terminal dividend, but limited by transparency

The approach of life insurance companies will be forced on it by prevailing customs in the country concerned.

Margins for future adverse experience

- Premium rates charged contain implicit or explicit margins to generate profits.
- Explicit margin – bonus loading
- Implicit margin – using conservative assumption in the pricing formula
- Due to in case of adverse future experience, company can repurpose those premium margins to absorb the shock
- In revalorization, the company distributes investment profit so any expense or mortality loss must be borne by the company. However, as these products are prudently priced, the likelihood of the company suffering large losses from this source is quite small.

- PH reasonable expectations may make it difficult for company to reflect the poor exp in bonuses
- Company cannot declare negative bonuses.

Business objectives of the company

- Main - Maximize profit distribution to PH to improve its competitive position but needs to be balanced with other objectives of the company.
- However, if bonus rates go downwards, it could expect to injure the company's competitive position and so lead to reduction in new business increasing per policy expense and marketing risk.
- However sometimes company have to reduce the rates to stay solvent and reflect where the current financials

4. Discuss the extent risk can be controlled under

- The conventional addition to benefit methods – smooth the bonus and retain the surplus for the years where the company's experience was worse. So that bonuses are unchanged
- Accumulating with profit – marketing risk if regular bonus is reduced by too much in any one year. Some smoothing is likely to be practiced but lesser than conventional. In case of terminal bonus, the company will have more scope to smooth and hence manage the risk.
- Revalorisation – company must declare the bonus according to surplus
- Contribution – similar to accumulating WP.

Policyholder's expectations

May have expectations regarding the profit distributions and the level of bonuses or dividends. Maybe built from : Past practice (bonus distribution) – general life insurance market practice – documentation issued by the insurance company (marketing literature, any projections implying a specific yield or eventual maturity benefit)

If PH realistic expectations are not matched, NB may drop and supervisory authority may intervene

5. Discuss the extent risk can be controlled

- The conventional addition to benefit methods – discretion to control bonuses
- Accumulating with profit – In theory, discretionary. However, PH may expect bonuses to be linked with underlying assets and hence market returns. This would not prevent the insurer from over declaring, but it might be difficult to under declare during the good times.
- Revalorisation – Actuary cannot influence bonus rates thus expectations should not be misled by literature or marketing , etc
- Contribution – similar to accumulating WP.

Provision of capital

- The pace at which profit is generated and profit is distributed might not be same
- If part of the profit is deferred to some future date before being distributed-company will have free assets-increase its ability to take on risk.
- This surplus could be used to support new business.

- Even in jurisdictions where reserves must be held for undistributed profit which is for future bonuses, it is generally the case that the overall reserves would be lower than had the profit been distributed.
- Where distribution is not being distributed as and when it arises, there will be years when the amount distributed exceeds the amount generated and vice versa.
- However, over time it would be expected that there should be a balance between the two. Sustained over-distribution could lead to an excessive drain on free assets.
- The extent to which it is possible to defer the distribution of profit depends on the form of the distribution.

Additions to benefits

In total amount of bonus, more in terminal=>more deferral distribution

In reversionary – super-compound will defer the distribution more than compounds which in turn defer more than a simple bonus. This is because the policies with 2 different bonus methods should result in the exact same total amount at the maturity date. Because super compound uses 2 different rates where bonus-on-bonus rates are amplified, the growth curve is incredibly steep. It acts like a rocket ship in the final year. So to hit the rocket shot the exact same final finish line as a slow and steady train, it has to start way behind the train the early years.

More benefits as terminal bonus, less guaranteed liability in monetary terms and more is discretionary in nature. The greater the amount of discretionary liability permits a greater defree of investment in higher-risk or higher expected return assets such as equities and properties

6. Distribution of profit will be more deferred for conventional with profit compared to accumulated with profit. Explain why?

- ⇒ This is because, in CWP, bonus is % added to the SA. This means that if a bonus is declared in Year 2, it increases a payout that is contractually locked away until maturity or death.
- However in AWP, the regular bonus is declared as a monetary interest rate credited directly to a Current Cash Account Balance. It will be hard to hold back too much, because it will make the bonus rate look small and lead to marketing risk.
- Because in CWP, allocations are tied to distant future obligations rather than a current account balance, the true economic distribution of that profit is heavily deferred.

In Revalorisation – less scope to defer

Contribution – some scope – can withheld some for terminal dividend – however this will be limited to policyholder's expectation on regular dividend each year -

General business environment (1)

Stakeholders and their impact on insurance market or insurer

Policyholder	Employers
Insurer	Investment management company
Distribution channel	Legal and govt

Regulatory and tax	Reinsurer
Competitors	TPA and other outsources

Professional guidance

Economic and other influence – inflation, business cycle, employment security, political stability

➤ **Policyholder**

- It is not compulsory to buy LI, therefore the likelihood of consumers purchasing depends on the combined forces of their own natural inclination to buy and the power of the insurer to sell.
- Power here implies – knowledge and information
- Customers are only expected to buy products which are useful to them, so companies have to sell products which just do that.
- Consumers would need knowledge on the benefits of LI and particular aspects of insurance companies and that's where the salesperson comes in.
- It helps when the product is easy to explain.

1. Life insurance is generally sold, not bought. Explain

- people may not be aware that they need the product. Could be because of – ignorance or belief that the event 'won't happen to them'.

Death may be a taboo subject

Products are difficult to understand

Difficult marketplace – the presence of many providers makes it hard for individuals to know which products or companies would be best to meet their needs

Need for advice before making long term financial commitments, which makes the whole process very time-consuming.

2. Outline the problems with using a salesperson for this purpose

- Problem with how the salesperson is remunerated, i.e., their commission
- Because a salesperson is rewarded for making sales depending on the type and size of product sold, not just giving advice and information.
- This could encourage them to sell more expensive or larger products, irrespective of keeping customer's needs in mind.
- do not meet the needs of policyholder, persistency risk, reputational and NB risk arises
- transparency of benefits at the time of sale is important
- Level of premium, guarantees offered and reviewability of premiums also adds on the likeness of the policy
- Legal requirements and tax benefits
- Innovative and what competitors are offering
- Underwriting process should be easy and fast
- Local culture impacts the purchase. Eg –
 - Degree of trust on financial services company
 - Level of education and financial awareness
 - The extent to which families provide support in tough times
 - The age at which children become self supporting

- Cultural attitude towards death.

➤ **Distribution Channels**

Insurance intermediaries

- Independence of any insurance company
- Sells different products of several different insurance companies to meet the needs and situations of their clients.
- Remunerated via commission payments by the companies whose product they sell or may receive fees from clients.
- They help the individuals that lack the time and expertise to find the products which are the best value and most appropriate for their needs by performing necessary research.
- However, these intermediaries are heavily influenced by their commissions.
- It will often be the client who will initiate the sale in case; however intermediaries will also promote themselves to existing or new customers. Promotions may include the mailshots of updates on either innovative products or tax regulatory change.

Ties agents

- Works on behalf of one or several life insurance companies.
- Only offer products of these life insurance companies.
- Typically they are employees of banks or some other financial institutions or individuals (in UK).
Adv of linking up with bank –
 - Give access to customers who already have established relationships with the company.
 - They can be reached via bank branches, mailshots and hence increase sales.
- With these agents more often the products they are selling are overlapped with each other.
- They are remunerated by the companies. Could be commission or salary + bonuses.

Own salesforce

- Employees of life insurance companies will only sell the products of that company.
- By commission or salary or both.
- Salesperson who initiates the sales, making use of client list or purchase leads (from marketing company).

Direct Marketing

Takes four main forms – mailshots, telephone marketing, press advertising, internet selling

For mailshots –

- get the list of names and addresses from its data or other affinity groups such as cc.
- will write to all of them to take out the policy.
- Company initiates the sales.

Telephone marketing –

- Both parties can initiate the sales

- Life insurance company calls – “call calling’

Press selling –

- Debatable who initiates the sales
- May include a short application form from the customers to complete and send in.
- May give telephone number or address from which further information and an application form may be obtained.
- May give a telephone number to call for the sale to be completed.

Internet selling –

- Limited to simple products to date.
- But can provide information leading to sales by telephone or any other means.
- Customers initiate the sales

The effect of different distribution channels

Demographic profile

- Different channels are likely to appeal to different people, according to their level of financial sophistication and level of income.
- Customers seeking product through
 - insurance intermediaries – financially sophisticated, wealthier - tend to seek out advice in managing their more complex financial affairs.
 - Major banks as tied agents – broad section of people with different levels of financial sophistication and income. It may still be upper level
 - Downmarket tabloid newspapers – least financially sophisticated - low levels of income however may indicate some level of education, and perhaps therefore relative wealth.
- Target market reached by distribution channels impacts the level of financial sophistication and income and mortality, withdrawal and morbidity experience.

Contract design

- Higher complexity for a more financially sophisticated client base.

Eg -

		Intermediaries	Own sales force	Telephone selling
Product	Term insurance	All products can be sold by them	Yes	yes
	annuities		Yes	Can be sold

	Unit linked flexible whole life		Can be sold	No
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- Annuities are rather complex because of many options and more complex policy conditions. There may also be tax and other legal requirements that have to be followed. Therefore, we may have something too complex to explain over a phone call.
- Unit links can also be sold through sales force.
- Unit linked flexible whole life is too complex to be sold over telephone calls.
- An insurance company using different distribution channels may therefore sell different versions of the same products, varying by channel.

Contract pricing

➤ Effect of demographic assumptions-

- Level of underwriting will depend on the choice of distribution channel.
- Some products, like annuities (except where terms offered are favorable to lives with impaired health), unit linked endowment with small SA on death, other endowment product (unless death benefit is substantially higher than the maturity benefit) requires little underwriting.
- WP endowment will require more underwriting compared to other WP products because PH are more interested in insurance protection than the investment potential.
- Whole life will have little underwriting if SA is small.
- Another reason for low underwriting is that lot of underwriting may cause barriers to sales.
- Therefore the level of underwriting will then be reflected in the demographic assumptions used for pricing.
- The socio-economic selection mentioned above will also impact these assumptions.
- Withdrawal rates are likely to be affected by the level of financial sophistication and whether it was the policyholder who initiated the sales.

3. Why is business from intermediaries likely to be subject to most underwriting?

- Intermediaries represent the interest of the clients, therefore insurance companies needs to be aware of possible anti-selection that intermediaries may encourage.
- If intermediaries 'initiate the sales' the risk of anti-selection increases.
- Customers will be of high net worth resulting in high cover.
- Prices of policies need to be competitive, which could be achieved by careful selection of risk through underwriting
- However underwriting cannot be too strict because it may work against the insurance company, because then intermediaries may discourage buying insurance from them.

4. Regular premium W/O P whole life insurance to cover funeral expenses marketed to manual workers.

Benefit in the first 12 months = premium paid. Comment on the mortality exp.

- Poor. The target population has poor health and there is no underwriting.
- Experience might be worse than general population mortality until the effect of anti-selection is diminished.

- Low death benefit in the first 12 months would at least discourage some attempts at anti-selection.
- Although 12 months may not be a sufficiently long period.

➤ Effect on the need for competitive terms

- Competitiveness of premium rates.
- Insurance intermediaries will recommend to their clients the companies with the most competitive rates, other things being equal (including commission)
- Tied up's product are generally competitive or they good damage the name
- The sales force will not be in a competitive situation.
- Press advertising in financial magazines or heavyweight newspapers may reach financially sophisticated costumers who are keep to compare rates.
- Competition through:
 - innovative features or attractive option
 - More complex products may be difficult to compare across companies
 - Investment performance of saving products
 - Level of customer service or admin support

5. NP term insurance and Unit linked endowment assurance are bought from insurance intermediaries.

Comment on the need for competitive rates/charges for these two contracts

- Term insurance – easy to understand and simple. High need for competitive rates. Can provide renewable or conversion options.
- Unit linked – expense and charges play important role

Distribution channel and the actuarial control cycle

- Decision on distribution channels to use.
- Distribution strategy to decide in order to meet the future profitability targets maintaining an acceptable level of risk.
- Analysis of the possible channels available in the market and those being used by the competitors of the company should indicate a likely successful market.
- The choice of distribution will impact number of following assumptions, such as:
 - The volume, type and mix of business likely to be sold including contract design, pricing and policy size.
 - Mortality, morbidity and withdrawal experiences in the new business.
- The strategy which aligns with the company's aim, according to modelling, might be adopted.
- The cost of each distribution channel should also be considered.
- Sales strategy should be continuously monitored to ensure the level, type and quality of new business is materialized as expected.
- Companies should always be striving to make improvements in sales distribution strategy.
- Analyse the experience of individual sales outlets and can point to areas of poor and good performance.
- A professional approach to sales distribution is completely necessary.
- Consumer's interest should be borne fully in mind.
- Unethical selling should be avoided.

In class notes –

- Target market based on financial sophistication, simple/complex products, affordability, socio-economic status, demographic mix (assumption experience)
- Different markets are targeted by different channels
- Level of risk/claims experience - Persistency
- Level of commission
- Training and qualification requirements
- Cost (admin, setup, training, commissions)
- Sales volumes
- Underwriting and sales process
- Competitive terms
- Who initiated the sales and disclosure
- Mis selling and reputation

The General business environment (2)

Type of expense and commission, including influence of inflation

- Expense – commission and management expenses.
- Initial commission – on acquisition of a new policy, Renewal commission – renewal premium
- Lapse – part of initial commission may be recovered. With the risk of non-recovery.
- Management expense – overhead expenses that incur regardless of the amount of the NB&IF.
- Overheads include, cost of general management, company's service department and accommodations.
- The company does expense loading in the premium or takes a separate fixed amount from each policy to cover the expenses. However, there is a risk that these loading will be insufficient to meet the actual expense incurred, it could also be due to inflation
- Many underlying costs are directly and indirectly linked to wage, salary levels, commodities
- NB could be cyclical

Economic environment (State of economy in which the insurer operates.)

The availability of asset types, and their short- and long-term expected yields, will determine how well an insurer can choose investment and probability of securing return assumed.

- Volatile economy – products are expensive and less purchased, require more capital to back the liabilities and cover the risk of future investment not being enough to cover the insured event. This results in high reserves thus low profit and thus reducing the value to SH and hence require higher premiums
- Insurer investing in risky assets for higher returns leads to a greater risk of the required return not being achieved.

Legal Environment (legal risks may arise from)

- inconsistencies between the policy document and any other relevant representations made by the company or its agents.
- Unfair contract terms voiding clauses of the contract.
- Some principles related to PH's reasonable expectations acting unfavourably to the insurer
- A compounding difficulty here is that insurance contracts span many years and are open to developing legal cultures, interpretations, and court judgments.

Regulatory restrictions and environment

To protect the insured from default of payment from the insurer. Common restrictions are

- Type of contract- authorized to write one or two classes of life insurance products, restrictions on the range of unit funds or a minimum level of death benefit.
- requirement will be different basis on saving and protection prod
- The premium rates or charges that can be used for some type of contract
- Rating Factors that can be used to determine the premium or discrimination charges eg – gender is not allowed.
- Terms and conditions of the contract offered, such as how to calculate SV or paid-up
- Customers and regulators in a number of countries have been concerned about poor value for money on early withdrawal from life insurance contracts. High initial expenses, including up-front commission, are responsible for this problem. One possible solution is for regulators to impose minimum values (for example specified percentages of premiums paid). Alternatively, where policy values are calculated on a prospective basis, the basis for the calculation could be specified or restricted.
- The Distribution channels:
 - minimum training requirements for insurance salespeople
 - the right of cancellation of contract by the policyholder
 - the illustration of possible maturity values and surrender values, perhaps on specified or restricted bases.
- the ability to **underwrite**, for example a prohibition on the use of the results of genetic testing, or prohibition of use of past claims history or medical history
- the amount of **NB** that may be written. In most countries there are regulations regarding the minimum level of mathematical reserves that must be held, often combined with minimum solvency capital requirements. These regulations have the effect of limiting the capital available within a company to write new business and effectively placing a minimum requirement on the finance required to write a contract.
- **Investment restrictions** are also to be expected, such as
 - the types of assets to invest – some risky assets can increase the yield of the company but such assets will not enable the company to maximise their expected returns.
 - the amount of any type of asset that can be considered for the purpose solvency
 - the extent to which mismatching is allowed at all - This could involve the possible setting up of an investment mismatching reserve. The more a company decides to invest in riskier

assets with a higher expected return, the higher could be any such resulting reserve. This would increase the value of the liabilities and reduce the available free assets.

- The regulatory environment is likely to have a significant effect on the design of the contracts, as the companies will want to make the best use of any regulatory opportunities available to them. Conversely, contract design will have to allow for any constraints imposed.

Regulations to reduce investment risk

- Restrict the type of asset
- Restrict the extent of investment in particular asset class eg, minimum or maximum % in low or high risk assets
- Limit investment in individual assets
- Introduce admissibility limits, i.e., restrictions on the amount of any type of asset that can be taken into account for the purpose of demonstrating solvency.
- Require investment mismatching reserves
- Limit the extent to which mismatching is allowed.
- Regulation could influence the choice of asset indirectly. Eg - by defining the methodology to calculate valuation DR.
- Similarly method use to value assets for regulatory purposes could influence asset choice
- IR can be reduced by increasing the strength of reserves required by introducing minimum reserve requirements.
- Introducing minimum solvency requirement against IR
- Tax rules could influence the relative attractiveness to different class of assets
- Restrict the investment guarantees that insurance company can offer

Regulations to reduce the investment risk taken by PH

Pre-Sale

- Introduce regulation on product design, for example - limiting the type of investment contract can be sold, restricting the range of investment funds that can be offered, setting maximum or minimum percentage that must be held in specific asset classes, minimum guarantees
- Use the tax system to incentivise/disincentivise types of contract according to their levels of investment risk.

Point of sale

- Introduce regulations to ensure the investment risks are highlighted to the customer. For eg - specifying information that must be provided, limiting the sales channel that can be introduced, specify the certain type of advice that must be provided

Post-sale

- Communication for example - regular statement providing measure of investment risk

Regulatory requirements for a policy to cover a key person

- Recruitment costs of a suitable replacement

- The cost of buying out the key person's shares/partnership stake.
- Protecting profits from losses due to delays in major projects in which the key person was involved.
- The cost of temporary cover while a permanent replacement is found
- Paying off any business loans in the name of key person.

Tax regime

May be taxed in different ways. Tax on-

- annual profit, i.e., the excess of the change in the value of the (assets over liabilities) or free assets. Calculated as $\text{Free asset} = \text{value of asset} - \text{liability} = (A1-A0)-(V1-V0)$. This approach focuses on the SH profit. And profit distributed to PH excludes the profit calculation
- investment income/ gains less than some or all of the operating expenses of the company. Tax rules will specify the type of investment income, gain (realized and not realized) and expenses (as they reduce the real return) included in taxation and any adjustment to be made.
- In addition, there may be tax on the premium income

Effect

- Within a country different types of life insurance business may be taxed by different methods.
- This means that it is at the lower cost for the consumer if certain forms of benefit.
- Tax treatment can make life insurance more or less attractive as a saving medium.
- Tax concessions may make the sale of certain types of contracts easier.
- The tax treatment of policy proceeds when they are received by PH can distort buying habits.
- Overall attractiveness of a life insurance product depends on –
 - o The taxation treatment of premium paid
 - o The taxation of the insurer's funds during the life of the contract
 - o The taxation treatment of eventual policy benefits.
- Therefore, to maximize the tax benefits, product design will be altered.

Risk - Taxation changes over time, so it is important to bear this in mind when benefits are guaranteed over the long term

Professional guidance

- Actuarial associations will often issue professional guidance for the actuaries on advising.
- These will typically give the framework of points that they need to consider in carrying out their responsibilities so as to maintain professional standards.
- However, these guidance do not restrict the actions and may even provide protection against pressure from proprietors to agree to a course of action that may not be in the best interests of PH.
- professional guidance on the interpretation of government regulations where the government does not want to be overly prescriptive in its requirements to restrict unduly the actions of life insurance companies. So the government will look to the actuarial profession to set limits.
- To the extent that professional guidance adds safeguards as to the proper running of life insurers

Supervisory reserves and capital requirements (1)

Background

Purposes of reserves

Reserves are calculated to

- demonstrate solvency to supervisory authority
- To investigate the true financial picture

Why solvency capital requirement

- Insurer stays solvent, i.e., able to meet guaranteed liabilities
- Hold some buffer over reserve in case of adverse events, i.e., prudent assumptions
- Buffer is not to be used elsewhere
- SCR is part of the Available capital
- Total asset = Reserve + Available asset

Reserve+ SCR should be sufficient ; Prudent reserve => small SCR or BEL reserve => significant solvency capital

Supervisory authority sets the 'minimum valuation' standard. This could be set of rules

To get the 'true' financial picture, i.e., value of asset and liabilities, assumption must be realistic. Used For internal management

- Decision making on how much NB can they take
- Administration of policy
- Financial management
- For WP – bonus/ dividend rates
- Profitability for SH and management

Gross premium valuation method

A method to place the value on a LI company's liabilities that explicitly values the future office premiums payable, expenses and claims, with the latter possibly including future discretionary benefits

Reserve = PV of future claims + PV of future expenses – PV of future premiums

Formula method

For Regular premium NP endowment assurance –

PV of sum assured + PV of renewal expense - PV of office premium

Where SA is Sum assured, RE is renewal expense and P' is office premium

- Usually, policies are priced such that PV of premiums exceed that of claims and expense. So above equation theoretically should be negative
- Initially, reserves are negative due to high initial expense, however any prudence can lead to it being +ve.
- Due to future expense inflation and cost of benefit payment will increase in future therefore the value of future benefits+expense > premiums. Giving a smooth progression of reserves.

Cashflow method

- Net cashflow is calculated for each future point in time.
 - PV of cashflows
 - Most valuable for unit linked
1. Advantages of cashflow method over formula
- o Identify whether net cash flows in any particular period are +ve or -ve
 - o Allow withdrawals easily
 - o Easily cope with complex charging and benefit structure
 - o Easily cope with charges and benefits which depend on future assumptions
 - o It is easy to incorporate assumptions that vary over time including stochastic assumptions
 - o Easier to allow for the impact of reinsurance
 - o Tax can be allowed more appropriately

Non – unit reserves

The structure of the unit linked contract is such that the company's liability is denominated partly in terms of units and partly in terms of non-unit.

Non unit liability includes - administration expense, mortality cost in excess of unit reserve. Withdrawal cost if guaranteed surrender value exceeds unit reserve, the future cost of other guarantees and policyholder options

This leads to the requirement of both unit and non-unit reserve (in the UK known as sterling reserve).

Unit reserve - reserve set in respect to unitised contracts. Representing liability in terms of units held under the contracts. = no of units * bid value

Non-unit reserve - PV of the excess of non-unit outgo (eg expenses, benefit in excess of the unit fund) over the non-unit income (charges, unallocated premiums). A discounted cashflow method is used.

If the PV of the non-unit income exceeds that of the non-unit outgo, then the overall value of reserve would be negative. It may or may not be permitted for a negative non-unit reserve to be held, depending on the regulatory regime

Prudential valuation

- Mathematical reserves are required to be prudent
- The non-unit reserve is typically defined as the amount required to ensure that company can pay the claims and meet its continuing expense without recourse to further finance.

To calculate the non-unit reserve it is necessary to consider the year by year incidence of the various components of the non-unit cashflows to determine if and when a non-unit reserve is required.

The company should project forward its non-unit cashflows on the reserving basis. This may need to be performed on a policy-by-policy basis. The non-unit reserve can then be calculated as follows:

- To calculate process starts with the last projection period in which the net cashflow becomes -ve.

- An amount is set up at the start of that period which is sufficient, allowing for earned investment return over the period, to 'zeroise' the negative cashflows
- This amount is then deducted from the net cash flows at the end of the previous time period.
- The process continues to work backward towards the valuation date, with each negative being 'zeroised' in this way.
- When the process has been completed, if the adjusted cashflow at the valuation date is negative then a non-unit reserve is set up equal to the absolute value of that negative amount.

Best estimate valuation

Non-unit reserve can be calculated for a best estimate valuation.

The calculation would value all future non-unit cashflows, i.e., it would not disregard cashflows occurring after the last projection period in which there is a net outflow, and there would be no other restrictions.

Negative non-unit reserves can be held. -ve reserve means that we are treating the contract as an asset (i.e., more money is going to come in through future charges than go out through future costs). This sounds a reasonable approach for a best estimate valuation as the company realistically will make a gain from the contract.

Negative non-unit reserves

- It may be permissible to hold a negative non-unit reserve under a contract where future non-unit income is expected to be more than sufficient to meet future non-unit outgo.
- -ve reserve represents 'loan' from other contracts which have positive non-unit reserve. The loan will be repaid by the emerging future profits from the policy for which the negative non-unit reserve is held

When will negative non-unit reserve be held?

Regulations may specify that:

- The sum of the unit and non-unit reserve for a policy should not be less than any guarantees SV.
- The future profits arising on the policy with the negative non-unit reserve need to emerge in time to repay the 'loan'
- After taking account of the future non-unit reserves, there are no future negative cashflows for the policy, i.e., there should be no future valuation strain.
- In aggregate, the sum of all non-unit reserves should not be negative. - This could be from anywhere in the company's business, not only from the unit linked policies.

Companies also do want to hold the negative reserve because they will reduce the total reserve under the contract and hence improve the capital efficiency of the product. They can be used whenever there are future positive cash flows that the company wants to take advance credit on. For example - initial expense are too high in a policy, creating negative cashflows, therefore advance credit can be taken to reduce the new business strain.

Calculating negative non-unit reserves

- Project the expected future non-unit cashflow from the policy

- Identify the last cashflow
- Set the reserve as an amount needed to meet that cashflow at that point in time. (even if cashflow is positive, set non-unit reserve as -ve)
- Check the total reserve (unit+non unit) is greater than the SV,
- Move back to the next previous cashflow, discount the reserve and subtract from the reserve the new cashflow at the earlier time period.
- Carry on repeating the process, working backwards over time to the valuation date.
- This will give the required non-unit reserve

[Reserve available + cashflow - reserve required]

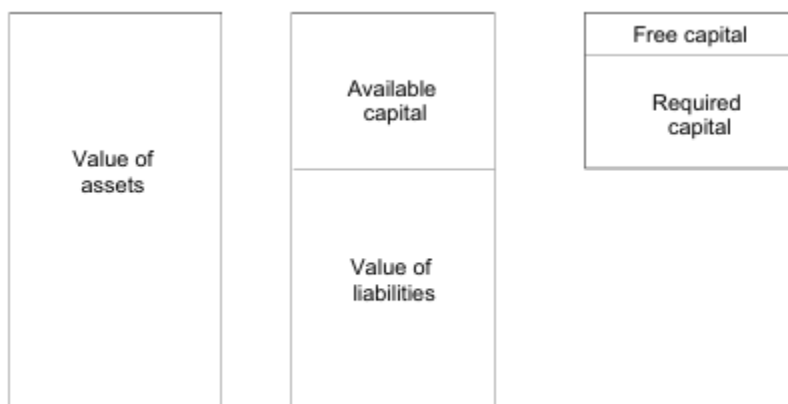
Features of the gross premium method

- Allowance for expense.
- Allowance can be made for bonus
- The future premium valued are the actual premium expected
- Any difference between the pricing and valuation basis will immediately can be taken as profit or loss
- Reserves may initially be negative for non-linked business, partly due to initial expenses and partly due to capitalising the expected future profit.
- The reserves tend to be quite sensitive to changes in basis.

Supervisory reserves and capital requirements (2)

Main concern of supervisory is to ensure insurance companies have sufficient assets to cover their liabilities. This could be achieved by:

- Insurance company hold reserve at prudence basis,
- Requiring minimum level of solvency capital in addition to reserves calculated at best estimate
- Requiring a combination of prudence reserve and solvency to be hold



Required capital is the minimum amount of capital that must be held to satisfy the supervisory authorities. So to be solvent free capital should be positive.

Passive valuation approach – reserves are relatively insensitive to each method

Reserving vs Pricing assumption

- Reserve – Reserve is real world assets, held to back future liabilities.
- WP contract – calculate premium rates using prudent assumption and using same assumption using supervisory purposes. Because surplus will then emerge from actual experience being better than that assumed in the prudent assumptions and that surplus is later distributed to PH as bonus and dividends. This approach is not feasible for WoP because they are priced with relatively small margins for competitive reason.
- In some countries, when actuaries' price policies use best estimates, they use high risk discount rates so that policy remains profitable because higher discount rate drastically reduces the PV of future cash flows forcing system to calculate a target premium that clears that high profit hurdle for SH.
- However, this high RDR (opportunity cost for investor, i.e., yield SH demand to justify locking their capital up in the insurance company rather than investing elsewhere) cannot be used for reserves, because if we do, then actuary is assuming that money will grow exceptionally fast and we need to hold very little cash to meet future claims. But if the investment earn less than they won't be able to meet liabilities.
- Thus shareholders' desire for profit does absolutely nothing to change the physical compounding power of the bonds sitting in the insurance company's vault
- Therefore, some regimes use market consistent assumption in base reserves.

Market consistent valuations

Market consistent methodology (also known as fair valuation)

- Market consistent value of a liability/reserve is the price that someone would charge for taking responsibility for the (i.e., ownership) of liability, in the deep and liquid market.
- However, insurance market is not deep and liquid, for MC valuation, we find a asset with similar cashflows in any other market, where that asset is traded in large number and large trade would not materially affect the prices, with same stream of cashflows
- Assumption – future IR and discount rates = risk free rates
- Risk free rates are determined based on govt bond yields or on swap rates (where swap market is deep and liquid).
- Since Govt bond and swap both have credit default risk, so there is a default premium (spread) in the Govt bond return and swap rate return. therefore, discount rate or risk free rate = Govt bond return/swap return – default risk premium
- Similarly, there is a liquidity risk and default risk for corp bond, which gets compensated with default premium and illiquidity premium., i.e., Corp bond return = RFR + default prem+ illiquidity prem Therefore, in case of corp bond,

$$\text{Risk free Rate} = \text{Corp return} - \text{Illiquidity premium} - \text{Default premium}$$

Illiquidity premium :

- Due to the addition of illiquidity premium, the discount rate will increase and the value of liabilities will decrease. Resulting in improved solvency position of the company.
- The use of an illiquidity premium is normally restricted to long term predictable liabilities for which matching assets can be held to maturity.
- In these circumstances the insurer is not exposed to the risks associated with the sale of illiquid assets and so may be permitted to increase the risk free discount rate accordingly.

- The liabilities under annuities are more predictable than other products as annuities
 - Cannot be surrendered,
 - Are paid regularly until the death of PH, or dependant
 - Do not pay a sum assured no death.
- If the company have large portfolio than this increases the predictability of the cashflows, which are based on the expected lifetime of the PH or dependent.
- Therefore assets can be held which match the expected duration of the annuity payments.
- For other products, where liability profile is less predictable, assets would typically not be held to maturity, so the insurer would be expected to the illiquidity risk of having the assets before maturity.
- Therefore the illiquidity premium will normally only apply to annuities.
- However, in some regulation regimes, regulators allow insurers to use a higher DR than RFR if certain conditions are met. Conditions are
 - Corp bonds used to back reserve/liability.
 - Nature of liabilities are long term and predictable - This condition is met when there is no surrender value in a contract, for example in immediate annuity when annuity amount is backed by the coupon payment of bonds. By liability being backed we mean, terms of bond, cashflows, timings of coupon payments, nature and currency should be the same as contract.
 - Bonds held to match the liability are held till maturity. (If they are held till maturity then liquidity risk will be 0, because then we are not selling the assets in the market within the term of bond)

But still we are exposed to default risk, therefore increased RFR = Corp bond rate + Illiquidity prem

- So, Total asset is looked at market value, i.e., cashflows are discounted back at-risk free rate irrespective of what they'll actually earn.
- All other risks that insurance companies are exposed to is considered in solvency on either prudent assumption or best estimate assumption.
- Best estimate assumptions are hedged by taking some margin, however still some risks are unhedgeable.
- That's where we use Risk Margin. So total reserve and risk margin represent the liability.

Risk Margin

- It may be difficult to obtain 'market consistent' assumption for some elements of the basis, such a mortality, persistency or expenses, for which there is not a sufficiently deep and liquid market in which to trade or hedge such risks. It is then likely a risk margin is included in respect to these assumptions.
- It is compensation required by 'market' in return for taking on uncertain aspects of liability cashflows.
- This could be done by applying margin to each individual assumption.
- Alternatively, an overall margin is calculated using the 'cost of capital' approach.
- This involves projecting the future capital the company is required to hold in respect of these risk, where this is determined at the end of each projection period during the run-off of the business. The projected capital is determined according to the relevant regulatory basis, and is the amount required to be held in excess of the projected liabilities at each period.
- These projected capital amount is then multiplied by a cost of capital rate. This rate can be considered to represent the cost of raising incremental capital in excess of the risk-free rate, or alternatively it

represents the frictional cost to the company of locking in this capital to earn a risk-free rate rather than being able to invest it freely for higher rewards.

- The cost of capital rate may be determined as the excess of the weighted average cost of capital over the risk free rate and so may be term dependent. In some calculations it is a specified fixed rate.
- The product of the cost of capital and the capital requirement at each future projection point is then discounted, using the market consistent discount rates, to give the overall risk margin:

$$\text{Risk margin} = \sum(k_t * C_t) / (1 + r_t) \text{ or } = \text{PV}[\text{run off SCR} * \text{Cost of capital rate}] \text{ at } t=0$$

Where k_t is the cost of capital 'charge' for the time t , C_t is the required capital at time t and r_t is the risk free interest rate for maturity t .

This involves

- For some this can be straightforward for other it might be complicated if the calculation required capital itself requires projections, stochastic modelling and/or application of correlation matrices.
- In such situations, a simplified approach can normally be used. This might involve selecting a driver (eg sum at risk or reserve) which has an approximately linear relationship with the required capital or its components.
- The initial capital requirement can be expressed as % of that driver, and the projected capital is then approximated as the same percentage of the projected values of the driver.
- In practice, more sophisticated methods using a combination of drivers and correlations may have to be used.

Solvency capital requirements

Purpose

Protect PH against -

- The reserve being underestimated and
- A drop in the value of assets

The level of SCR may be specified as a formula or it may be based on a risk measure such as VaR.

Interplay between reserves and solvency capital requirement

- Adequacy of solvency capital requirement and reserves cannot be looked in isolation.
- Practice on the relative balance between the two components varies between the countries and regulatory jurisdictions.
- Reserves are set up on best estimate, i.e., with relative small margins, then SCR is substantial and is determined by using risk-based capital techniques.
- The aim of the risk based is to set aside the extra amount of capital where the amount is appropriate to the extent of the risks involved.
- In other countries, reserves are set up on prudent basis with small SCR which is not specifically related to the risks borne by the company.

VaR approach

- Example of risk based capital approach, normally expresses at a minimum required confidence level (eg 99.5%) over a defined period.
- So the insurer calculates the amount of capital it needs so that its assets will exceed its liability in one year's time with the regulatory defined confidence interval (say 99.5%) or they may be determined by the risk appetite of the company..
- The supervisory balance sheet is subject to stress test (or 'shocks') on each of the identified risk factors, at the defined confidence level and over the defined period.
- This balance sheet will typically be on a market consistent approach.
- Examples of such shocks are - mortality, interest rates movements, credit risk, etc.
- The (market consistent) surplus is then recalculated at the end of the period.
- The shocked surplus at the end of the period is then discounted back to time zero using the appropriate discount rate.
- Applying stress tests to each different risk factor gives a capital requirement for each separate risk in isolation.
- These separate stresses need to be combined to give an aggregate capital requirement reflecting any diversification benefits that exist between the risks (i.e., the degree to which individual risks are correlated).
- This may be done using the correlation matrix or copulas.

$$\text{Aggregated capital requirement} = \sqrt{\sum_i \sum_j \text{Corr}_{ij} \times \text{Cap}_i \times \text{Cap}_j}$$

where:

Cap_i = capital requirement under risk i

Corr_{ij} = correlation between risks i and j .

j i	Mortality	Lapse	Expense
Mortality	100%	0%	50%
Lapse	0%	100%	50%
Expense	50%	50%	100%

Question

Calculate the aggregated capital requirement using the table above, if the separately calculated capital requirements are as follows:

- capital requirement for mortality risk = £100 million
- capital requirement for lapse risk = £10 million
- capital requirement for expense risk = £20 million.

Solution

$$\begin{aligned}
 \text{Aggregated capital requirement} &= \sqrt{100^2 + 10^2 + 20^2} \\
 &\quad + 2 \times 0.5 \times 100 \times 20 \\
 &\quad + 2 \times 0.5 \times 10 \times 20 \\
 &= £112.69 \text{ million}
 \end{aligned}$$

- Correlation matrices may be specified by the regulator.
- Under the extreme event conditions being tested, correlation may differ from those observed under 'normal conditions'.
- The aggregate capital requirement allowing for diversification will usually be less than the sum of the capital requirements for individual risks.
- A combination of a certain subset of events happening at the same time, with an overall probability level of 1-in-200, may produce a higher capital requirement than combining all of the individual capital requirements for separate 1-in-200 events using a correlation matrix.
- This is caused by the 'non-linearity' and 'non-separability' of risks.
- Non-separability refers to the ways in which risk drivers interact, i.e., if two events happen together then the combined effect is worse than the individual effect.

(Linearity means that the capital required is a linear function of the risk drivers).

- Allowances need to be made in the VaR calculation for these effects.

- Typically, stochastic models are used to quantify the capital requirements in relation to economic risk.
- The probability distribution used should properly reproduce the more extreme behaviour of the variable being modelled, both in the size of the tail of the distribution and in the path taken during the simulation period
- The real world asset model would be used and should be arbitrage free.
- These models are usually calibrated to actual historic values, but advanced techniques may be required to ensure appropriate fit to the tail of a distribution to ensure that the frequency of extreme outcomes is not understated.
- Other possible method of calculation may exist, such as 'run off' method which looks at the amount of capital needed at outset to ensure a firm's ability to cover its liabilities until the last policy has gone off the books, allowing for suitable stresses to the risk factors.

Active and passive valuation approaches

Passive valuation approaches

- Valuation strategy insensitive to market condition
- Valuation basis is updated very infrequently.
- Example will be the net premium valuation approach for the liabilities which is fairly insensitive to the net premiums also being recalculated under the new assumptions.

Net premium valuation - PV of contractual liabilities - value of the future net premiums allowing in each case only for mortality and interest. It ignores expense. Net premium is the premium calculated on the valuation assumptions and payable under the same conditions as the office premium, that will provide the contractual benefits offered at the commencement of the policy. The key point here is that the calculation makes no reference to the actual premium paid.

$$S.A_{x+t:n-t|} - P.\ddot{a}_{x+t:n-t|}$$

where: S = sum assured

$$P = \text{net premium} = S \cdot \frac{A_{x:n|}}{\ddot{a}_{x:n|}}$$

x = age at outset

n = contract term

- Under passive approaches, assumption may be 'locked in', i.e., they remain unchanged from those used when the policy was first written and the liability for it first determined.
- However, non-economic assumptions are updated if experience worsens, in order to recognise the related loss and the need for higher reserve at that time.
- For the valuation of assets, an example of a passive approach would be the use of historic cost or 'book value', possibly with amortisation (or 'write-down') over time.

- By using book value we are ignoring the impact of changes in market prices. This might be suitable if the valuation of liabilities is also largely ignoring market conditions. Eg if the valuation interest rate is locked in or if a net premium valuation is used.
- Solvency capital requirements may be determined using a simplified approach such as holding a prescribed percentage of mathematical reserves.
- Passive valuation approaches: tend to be more straightforward to implement ; involve less subjectivity; (to the extent that they are used for accounting principle) results in relatively stable profit emergence.
- Key disadvantages - risk of becoming outdated; insensitive to market changes and valuation basis updated infrequently resulting in ignoring important trends eg - rising inflation. Therefore there is a risk that a passive approach provides a false sense of security. Management may fail to take appropriate actions in response to emerging problems until too late, because the solvency position hasn't appeared to change.

Active valuation approach

- Based more closely on market conditions, with the assumptions being updated frequently.
- Eg - market consistent approach for Asset and liabilities and a risk based approach for SCR
- It is more informative in terms of understanding the impact of market conditions on the ability of the company to meet its obligations. Particularly in relation to financial guarantees and options.
- However, results are more volatile under such approaches.
- Under adverse equity market conditions, an active valuation approach would indicate that higher capital requirements.
- In order to reduce this, companies would need to sell equities - which itself could exacerbate the market conditions.
- There is also a systematic risk, as this would be the case for all life insurance companies at the same time. Therefore it may be the case that regulators include amendments to the valuation approaches under such conditions, to avoid this situation.
- Dis - complex. So calculation can take longer to perform and more costly

Combinations

- The overall valuation approach may instead be somewhere between the two extremes including elements of each.
- This can result in greater mismatch between the assets and liabilities, and hence greater profits/losses or changes in free surplus when market conditions change.

Implication of changing from passive valuation approach to active valuation

Practicalities -

- Need to redesign their liability valuation system.
- These changes will incur costs: staff training, system changes
- Higher ongoing valuation cost will also be incurred
- Life companies will need to change their analysis of surplus processes and conduct more frequent experience analysis.

Overall valuation approach

- In addition to reserve for liabilities, companies will also need to consider how the method of asset valuation might change.
- The SCR will be more risk based
- This will result in more useful management information and informed decisions
- Companies will need to amend their models and costs.

Transition

- At the time of implementation, there will be one-of changes in life insurance company's solvency result
- This will need to be explained to SH and analysts.
- Changes in companies solvency position will also need to understand by regulators.
- At the point of implementation, valuation on both the existing passive approach and the new active approach may be needed.
- Consider the timescale for implementation of the proposal

Impact

- As a result of their solvency results, companies may need to raise additional capital
- Companies result will become more volatile
- Companies may therefore change their investment strategies to control volatility
- This could lead to sell off certain assets
- Companies may change their products to reduce volatility.